

Nguyễn Đại Đồng

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in dong-nguyen-hcmut 🔗 Noname-29-lnin

Education

VNU - Ho Chi Minh City University of Technology

Oct 2022 – Present

Bachelor of Science in Electronics Engineering

- **GPA:** 3.2/4.0 ([This link](#) 🔗)
- **Project 1:** Design and Implementation Viterbi Decoder Algorithm on FPGA.
- **Project 2:** Hardware Accelerator for Mean-Based Large-Scale Sorting using PIM Architecture.

Research Interest

Computer Architecture, RISC-V Processors, Digital IC Design, DSP on FPGA, Applied Electronics.

Technical Skill

- **Languages & Technologies:**
 - + **Hardware Description (HDL):** SystemVerilog, Verilog.
 - + **Software Languages:** C, C++.
 - + **Scripting & Automation:** TCL, Bash, Makefile.
 - + **Tools & Version Control:** Git, Neovim, L^AT_EX.
- **EDA Tools & Environments:**
 - + **Simulation & Verification:** Cadence Xcelium, Siemens QuestaSim, ModelSim, Verilator.
 - + **Synthesis & Implementation:** Cadence Genus, Yosys, Intel Quartus (FPGA).
 - + **Custom IC/Layout:** Cadence Virtuoso.
 - + **Operating Systems:** Linux (Ubuntu, CentOS), Windows.
- **Hardware Skills:**
 - + **Digital Design:** RTL development, FPGA Prototyping, Verification.
 - + **PCB Design:** KiCad, Altium Designer (Basic).
 - + **Lab Equipment:** Oscilloscope, Logic Analyzers.

Projects

Design and Implementation Viterbi Decoder Algorithm on FPGA

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[/Viterbi_decoder](#) 🔗

- This project involves the design and implementation of a **Viterbi decoder** for convolutional codes with parameters (3, 1, 2) (constraint length $K = 3$, code rate 1/2) on an FPGA. The Viterbi algorithm is optimized for error correction in noisy communication channels, leveraging trellis-based path metrics and survivor path selection. The decoder is structured using key functional

blocks, including the *Branch Metric Unit (BMU)*, *Add-Compare-Select Unit (ACSU)*, *Path Metric Unit (PMU)*, and *Survivor Path Memory Unit (SPMU)*.

- To test the decoder, additional modules such as a *Parallel-In-Serial-Out (PISO)* register, *Serial-In-Parallel-Out (SIPO)* register, and *LCD IP core* (for hardware-based LCD control) are integrated. These components enable practical verification of the Viterbi decoder's functionality. The final implementation is deployed on a *DE2 development kit*, where the design is synthesized and programmed onto the FPGA for real-world validation.
- **Tools & Language:**
 - **Simulation:** Verilator, GTKwave.
 - **Implementation:** Intel Quartus Prime.
 - **Language:** SystemVerilog, C.

Hardware Accelerator for Mean-Based Large-Scale Sorting using PIM Architecture

[Noname-29-Inin/Sort_FPGA](#) 

- Designed and implemented the hardware architecture for a high-throughput sorting framework, based on the research "*A General Framework for Sorting Large Data Sets Using Independent Subarrays of Approximately Equal Length*".
- The RTL implementation targets performance optimization for large-scale datasets (from 2^{18} to $2^{32}+$ elements), supporting multiple data types.
- Engineered the core processing unit to handle independent subarrays, optimizing for either area (serial processing) or throughput (parallel processing) to balance resource utilization and sorting speed on the FPGA.
- **Tools & Technologies:**
 - **Simulation:** QuestaSim, Xcelium, Simvision.
 - **Synthesis & Implementation:** Intel Quartus Prime, Genus.
 - **Languages:** SystemVerilog, C++, Python.

Other Projects

IEEE 754 FPU Design: FPGA Prototyping and Synthesis using FreePDK 45nm

[Noname-29-Inin/FPU_IEEE754-45nm](#) 

Thermistor-Based Temperature Measurement Circuit

[Noname-29-Inin/Temp_Circuit](#) 